Project Ideation| AI/ML Geek Room

NoteWise

Turning Hinglish lectures into clear, organised notes



Listen in class. Read your notes after.

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The Problem & Who Faces It

The problem

In many Indian college classrooms, teachers explain in Hinglish, switching between Hindi and English in the same sentence. Students must listen, understand and write at once, and a live lecture can't be rewound. Note-taking takes more mental effort in a second language. Miss one line and the next few make less sense, and a 50-minute recording is too long to revise from. Most speech-to-text tools are built for clean English or pure Hindi, not this mix.

A TYPICAL CLASSROOM SENTENCE

Toh yahan pe pointer ko dereference karenge, matlab address pe jo value hai woh mil jayegi.

● Hindi ● English

Who faces it

Students in Hinglish classes
Anyone trying to listen and write at once.

Students who miss a lecture
Illness, clashes, travel: no good way to catch up.

First-years and new learners
Fast speech plus new jargon is a tough mix.

Students who need extra support
Slow writers, hearing difficulty, ADHD or dyslexia.

Teachers and coaching classes
Could share clean notes without extra effort.

Why It's Worth Solving

≈150 vs ≈30

words per minute: a lecturer speaks, a student writes. We only catch about 1 word in 5.

250M+

people in India are estimated to mix Hindi and English when speaking.

4.3 crore

students in Indian higher education (AISHE 2021-22). Even a small share is huge.

If we don't fix it

Time

Hours go into rewriting notes, borrowing a friend's, or rewatching recordings.

Money

Students fill the gaps with paid notes, tuitions or coaching.

Learning

Small gaps pile up. A 2023 study found lecture summaries helped students score higher on quizzes.

Accessibility

Slow writers, students with hearing difficulty and those less at ease in English fall behind first.

Proposed Solution

Record a lecture, get structured notes with key terms soon after class.

01

Record

Phone mic in class, or upload an audio file of a lecture or meeting

02

Transcribe

Hinglish speech to text with timestamps

03

Clean up

Fix subject terms, remove fillers, split by topic

04

Make notes

Headings, definitions, formulas, summary, exam flags

05

Revise

Search, flashcards and practice questions

THE TEACHER SAYS

Toh yahan pe pointer ko dereference karenge, matlab address pe jo value hai woh mil jayegi.

● Hindi ● English

THE STUDENT GETS

Pointer dereferencing

Reads the value stored at the address a pointer holds.

★ Exam-important | Key term: dereference | ▶ 12:42

Every note links back to the exact moment in the recording, so students can check it.

Where AI/ML Comes In

Needs AI/ML

Hinglish speech to text

CORE

Speech recognition. Rules can't do this: the model must follow Hindi–English switching inside one sentence.

Audio clean-up

ANY OS SOFTWARE

Noise reduction and separating the teacher's voice: signal processing plus ML

Cleaning text and writing notes

CORE

NLP plus a subject glossary fix technical terms; an LLM writes summaries, key terms and quiz questions from the transcript.

Finding what matters

Using Database

Classification and clustering: split the lecture into topics, spot “this is important” moments.

Types of AI/ML needed

CORE

- Speech recognition
- NLP
- Generative AI / LLM
- Signal processing

Using Database

- Classification
- Clustering
- Computer vision
- Recommendation

No AI needed: recording, storage, search, app screens and PDF export. That is normal software.

What Data Would We Need?

Data types

Audio

Real Hinglish lectures from different teachers, accents and noisy rooms (with permission).

Transcripts

Human-checked text for those recordings, to measure word error rate.

Course text

Slides, syllabus and an ECE glossary to catch technical terms.

Feedback

Which notes students edit or mark wrong, to keep improving the system.

Where could it come from?

MUCS 2021 (OpenSLR 104)

About 90 hours of Hindi–English tutorial speech. Closest to lectures I found.

IITG-HingCoS and HiACC

Public Hinglish speech sets (HingCoS is about 25 hours), mostly not classroom lectures.

AI4Bharat datasets

IndicVoices (~12,000 hours of Indian speech) and Vistaar benchmarks for testing.

Does Something Similar Exist?

Speech-to-text models

Whisper · Sarvam Saaras v3 · Shunya Labs Hinglish Whisper

GOOD AT

Speech to text. Saaras v3 even has a “codemix” mode for Hinglish.

GAP FOR US

Just a transcript, no notes. Plain Whisper had a ~34% word error rate on read Hinglish vs ~8–16% for newer models, and can invent text in silence.

Meeting note-takers

Otter.ai · Notta · HappyScribe

GOOD AT

Live transcripts, speaker labels, AI summaries. HappyScribe even detects Hindi–English switches.

GAP FOR US

Made for meetings, not lectures. Nothing on Hinglish classrooms in what I found, and Otter supports only a handful of languages.

AI lecture-note apps

Audionotes · TLDL

GOOD AT

Record a class, get notes, flashcards and quizzes. Audionotes lists Hindi among 80+ languages.

GAP FOR US

Generic. I found no Hinglish tuning, noisy-classroom handling or subject glossary.

What's Different? NoteWise

Made for Hinglish classrooms

Handles language switching mid-sentence, plus fans, chatter and a far-away phone.

Notes made for studying

Topics, key terms, formulas, exam flags and a quick quiz, in English, Hinglish or Hindi.

Fixes subject words

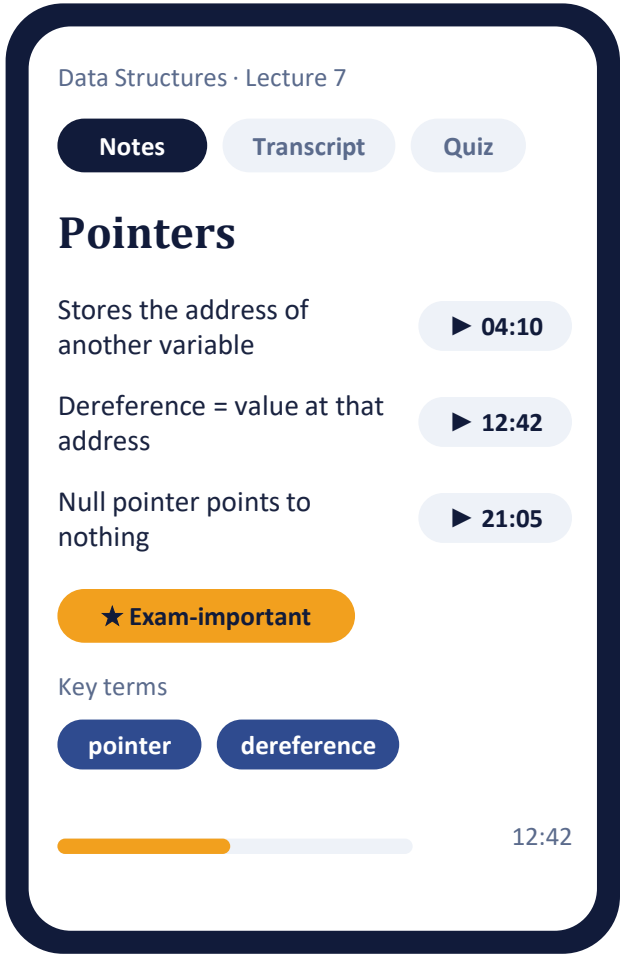
Upload the syllabus or slides once; a glossary fixes terms like “MOSFET” or “dereference”.

Tap a note, hear the teacher

Each note line links to its audio, so wrong notes are easy to spot and fix.

MY APPROACH

Don't rebuild speech recognition. Test 2–3 existing engines on real classroom audio, then put the effort into the notes layer and noisy rooms. First prototype: a web page where you upload a recording and get notes back.



Final product: mobile app + web dashboard

Challenges & Open Questions

Code-mixed accuracy

Language switches mid-sentence and Hindi has many Roman spellings. Even top systems miss about 1 in 8 words.

Noisy classroom audio

Far-away phone, fans, chatter, spoken formulas. Whisper-type models can invent text in silence, so add voice activity detection.

Data availability

Few public Hinglish lecture datasets (MUCS is ~90 hours). Labelling takes long and needs permission.

Made-up notes

An LLM may add things the teacher never said. Idea: link every note line to its audio and test with quiz questions.

Privacy, consent and cost

Voices are personal data (India's DPDP Act, 2023). Needs consent and delete options, plus GPU or API cost.

Will students tune out?

If notes are automatic, some may switch off in class. Maybe add quick recall questions after the lecture.

Sources & Research

The problem and students

Lecture speed and note-taking (Applied Cognitive Psychology)

Typing vs handwriting speed in lectures (Charles Sturt University)

Hindi–English code switching in Indian classrooms (NCERT, 2016)

AISHE 2021-22 key highlights (Ministry of Education)

Auto-generated lecture summaries and quiz scores (BEA 2023)

Fluency and cognitive effort in L1 and L2 note-taking (Piolat et al.)

Data and benchmarks

MUCS 2021 Hindi–English code-switching dataset (OpenSLR 104)

IITG-HingCoS: Hindi-English code-switching corpus

HiACC: Hinglish adult & children code-switched corpus

Hindi-English code-switching speech corpus (Ganji et al.)

AI4Bharat Vistaar benchmarks

AI4Bharat IndicVoices dataset

Hinglish speech technology

Tara model card: Hinglish benchmarks (Trelis Research)

whisper-hinglish-preview: Whisper vs newer models

Sarvam Saaras v3 output modes, incl. codemix

Whisper hallucinations on non-speech audio (arXiv 2501.11378)

Why speech recognition fails on Hinglish (Gnani.ai)

Shunya Labs Hinglish ASR model (Hugging Face)

Existing tools and privacy

Audionotes: AI lecture note taker

TLDL: AI Lecture Note Taker (App Store listing)

Otter.ai alternatives and language limits (Intermind)

HappyScribe AI notetaker for Hindi

India's DPDP Act 2023 brought into force (Hogan Lovells)